Week 8 - Convection and SALR: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Use $\Gamma_d = 9.8\ \mathrm{K\,km^{-1}}$ for the dry adiabatic lapse rate. Where a simple numerical value is required, use a representative saturated adiabatic lapse rate $\Gamma_s = 6.0\ \mathrm{K\,km^{-1}}$; remember that the true SALR varies with temperature and pressure. For dry potential temperature use $\theta = T(p_0/p)^{R_d/c_p}$ with $R_d/c_p = 0.286$ and $p_0 = 1000$ hPa.

YOU SHOULD TRY Question 1: Quantify the effect of latent heating on a rising parcel

Two parcels start at 20°C and rise 2.0 km. Parcel A remains unsaturated throughout and cools at the DALR. Parcel B is saturated throughout and, for this exercise, cools at $6.0\ \mathrm{K\,km^{-1}}$.

1. Calculate the final temperature of each parcel.
2. By how much warmer is the saturated parcel after the ascent?
3. Identify the physical process responsible for the difference.
4. Give one reason why the true saturated adiabatic lapse rate is not a universal constant.

IN CLASS Question 2: Lift a parcel through cloud base

A surface parcel has $T = 25°\mathrm{C}$ and $T_d = 17°\mathrm{C}$.

1. Estimate its LCL using $z_{LCL} = 125(T - T_d)$ m.
2. Estimate the parcel temperature at the LCL using the DALR.
3. If the parcel then rises to 3.0 km and cools at $6.0\ \mathrm{K\,km^{-1}}$ above the LCL, estimate its temperature at 3.0 km.

IN CLASS Question 3: Classifying atmospheric stability

For each environmental lapse rate below, classify the atmosphere as absolutely stable, conditionally unstable or absolutely unstable. Use $\Gamma_s = 6.0$ and $\Gamma_d = 9.8\ \mathrm{K\,km^{-1}}$.

1. $\Gamma_{env} = 4.0\ \mathrm{K\,km^{-1}}$
2. $\Gamma_{env} = 7.0\ \mathrm{K\,km^{-1}}$
3. $\Gamma_{env} = 11.0\ \mathrm{K\,km^{-1}}$

For each case, state what happens to an air parcel after a small upward displacement.

YOU SHOULD TRY Question 4: Potential temperature

An unsaturated air parcel has $T = 280$ K at 800 hPa.

1. Calculate its dry potential temperature.
2. A second parcel at the same pressure has $\theta = 304$ K. Which parcel is warmer at 800 hPa?
3. State why potential temperature is useful for dry adiabatic motion.

IN CLASS Question 5: Find an inversion and diagnose its consequences

A morning temperature profile is

Height (m)	0	200	500	800	1200
Temperature (°C)	5	3	7	5	2

1. Between which two levels is there a temperature inversion?
2. Calculate the temperature change across that inversion layer.
3. Would vertical mixing through the inversion be encouraged or suppressed?
4. A city lies below the inversion under light winds. Predict the effect on near-surface pollutant concentration and explain why.

EXTRA CHALLENGE Question 6: A simple Foehn-wind calculation

Air approaches a mountain at sea level with temperature 20°C. Its LCL is at 1.0 km and the summit is at 3.0 km. Assume it cools at the DALR below the LCL and at 6.0 $\mathrm{K\,km^{-1}}$ while saturated above it. On the lee side, assume condensed water has precipitated out and the descending air warms at the DALR all the way to sea level.

1. Calculate the temperature at the summit.
2. Calculate the temperature on the lee side at sea level.
3. By how many degrees is the lee-side air warmer than the original windward surface air?
4. Explain why the descending air is relatively dry.

EXTRA CHALLENGE Question 7: Identify the lifting mechanism from the weather situation

For each situation, name the main mechanism that initiates ascent.

1. Strong afternoon heating produces rising thermals over a dark land surface.
2. A moist westerly wind encounters the Pennines.
3. A cold front advances beneath warm air.
4. Surface winds from several directions meet along a narrow boundary.

For one of the four cases, explain why ascent does not automatically guarantee deep convection.

YOU SHOULD TRY Question 8: Use stability to predict cloud form

Two saturated layers have the following environmental lapse rates:

- Layer A: 3.0 $\mathrm{K\,km^{-1}}$
- Layer B: 8.0 $\mathrm{K\,km^{-1}}$

Use $\Gamma_s = 6.0\ \mathrm{K\,km^{-1}}$.

1. Which layer is stable with respect to saturated vertical displacement?
2. Which layer is unstable with respect to saturated vertical displacement?

3. If cloud forms in each, which layer is more likely to favour horizontally extensive layered cloud, and which is more likely to favour vertical cumulus development? Explain from parcel-environment temperature differences.

EXTRA CHALLENGE Question 9: Parcel versus environment through a conditionally unstable layer

At 1.0 km, a saturated parcel and its environment are both at 12°C. Between 1 and 3 km the environment cools at 7.5 $\mathrm{K\,km^{-1}}$ while the parcel cools at 6.0 $\mathrm{K\,km^{-1}}$.

1. Calculate parcel and environmental temperatures at 3 km.
2. Which is warmer, and by how much?
3. What does this imply about the parcel's buoyancy tendency?